(SoIM) BIG DATA TECHNOLOGIES & CLOUD-BASED ANALYTICS

Programme: PGDM
Trimester: IV
Total Contact Hours: 18 Hours
Course Orientation: Management, Business & Humanities
Level: Postgraduate Managemenr

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1. Course Description

Big Data and Cloud-Based Analytics have become important components of modern business decision-making. Managers today increasingly work with information generated through customers, employees, markets, digital platforms, social media, transactions and organizational systems.

This course introduces management students to the **fundamental concepts, applications and managerial implications of Big Data and Cloud-Based Analytics** without requiring advanced programming or technical expertise.

The course focuses on how managers can understand data, ask meaningful business questions, interpret analytical insights, support evidence-based decisions, and recognize the ethical and organizational implications of data-driven management.

The emphasis is therefore on **business understanding, managerial judgment, interpretation and application**, rather than on coding or complex technical implementation.

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2. Course Objectives

By the end of the course, students should be able to:

1. Understand the basic concept and evolution of Big Data.
2. Explain the major characteristics of Big Data in simple managerial terms.
3. Understand how organizations collect, store and use large volumes of data.
4. Explain the basic concept of Cloud Computing and Cloud-Based Analytics.
5. Identify business applications of Big Data across different management functions.
6. Understand how analytics can support managerial decision-making.
7. Interpret basic analytical outputs and translate them into business insights.

8. Understand the role of dashboards, visualization and data storytelling in management.
9. Recognize the ethical, privacy and social implications of data-driven organizations.
10. Develop a balanced understanding of the opportunities and limitations of Big Data and Analytics.

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3. Course Learning Outcomes

After completing the course, students will be able to:

CLO1: Describe the basic concepts and characteristics of Big Data.

CLO2: Explain the role of Cloud Computing and Cloud-Based Analytics in contemporary organizations.

CLO3: Identify appropriate business applications of data and analytics across management functions.

CLO4: Interpret basic data visualizations, dashboards and analytical findings for managerial decision-making.

CLO5: Evaluate the organizational, ethical, privacy and human implications of data-driven decision-making.

CLO6: Communicate data-based business insights in a clear and managerially relevant manner.

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4. Course Structure – 18 Hours

Module	Topic	Hours
I	Understanding Big Data in the Business World	3
II	Big Data Technologies – A Managerial Perspective	3
III	Cloud Computing & Cloud-Based Analytics	3
IV	Business Analytics & Managerial Decision-Making	3
V	Applications of Big Data Across Management Functions	3
VI	Data Ethics, Privacy, People & the Future of Data-Driven Organizations	3

Total	**18 Hours**

MODULE I – UNDERSTANDING BIG DATA IN THE BUSINESS WORLD

3 Hours

1.1 From Traditional Data to Big Data

- What is data?
- Structured and unstructured information
- Traditional databases versus Big Data
- Why organizations are generating more data than ever
- Data as a business resource

1.2 Understanding the "5 Vs" of Big Data

- Volume
- Velocity
- Variety
- Veracity
- Value

1.3 Sources of Organizational Data

- Customers and transactions
- Websites and mobile applications
- Social media
- Employees and organizational systems
- Sensors and connected devices
- Market and external information

1.4 Why Big Data Matters to Managers

- Understanding customers
- Identifying trends
- Improving operations
- Managing risk
- Supporting strategic decisions

Managerial Discussion:
"Does more data always lead to better decisions?"

Illustrative Cases: Retail, Banking, Healthcare, E-commerce and Social Media.

MODULE II – BIG DATA TECHNOLOGIES: A MANAGERIAL PERSPECTIVE

3 Hours

This module provides a **non-technical overview** of the technology ecosystem behind Big Data.

2.1 Understanding the Big Data Ecosystem

- Data generation
- Data collection
- Data storage
- Data processing
- Data analysis
- Data visualization

2.2 Databases and Modern Data Storage

- Relational databases
- Non-relational / NoSQL databases
- Data warehouses
- Data lakes
- Why different forms of data require different approaches

2.3 Introduction to Big Data Platforms

- Hadoop – basic concept
- Spark – basic concept
- Distributed computing – simple managerial explanation
- The idea of processing data at scale

2.4 From Data to Insight

Data → Information → Analysis → Insight → Decision → Action

Managerial Perspective

Students are not expected to build Big Data systems. Instead, they should understand:

- What these technologies do
- Why organizations use them
- What business problems they solve
- What questions managers should ask technology teams

Class Activity:
Students map the journey of customer data from an online purchase to a management dashboard.

MODULE III – CLOUD COMPUTING & CLOUD-BASED ANALYTICS

3 Hours

3.1 Understanding Cloud Computing

- What is Cloud Computing?
- Why organizations moved from traditional infrastructure to the Cloud
- Cloud versus traditional IT infrastructure
- Scalability, flexibility and accessibility

3.2 Basic Cloud Service Models

- Infrastructure as a Service – IaaS
- Platform as a Service – PaaS
- Software as a Service – SaaS

3.3 Cloud Deployment Models

- Public Cloud
- Private Cloud
- Hybrid Cloud

3.4 Cloud-Based Analytics

Understanding how organizations can use Cloud platforms to:

- Store data
- Process information
- Build dashboards
- Conduct analytics
- Collaborate across locations
- Support real-time decision-making

3.5 Business Benefits and Challenges

Benefits

- Scalability
- Cost flexibility
- Accessibility
- Collaboration
- Faster deployment

Challenges

- Security
- Privacy
- Vendor dependence
- Data governance
- Business continuity

Class Discussion:
"Should an organization keep all its data in the Cloud?"

MODULE IV – BUSINESS ANALYTICS & MANAGERIAL DECISION-MAKING

3 Hours

4.1 Understanding Business Analytics

- What is Analytics?
- Data versus information versus insight
- Descriptive Analytics
- Diagnostic Analytics
- Predictive Analytics
- Prescriptive Analytics

4.2 Analytics and Management Decisions

How analytics can support:

- Strategic decisions
- Marketing decisions
- Financial decisions
- HR decisions
- Operations decisions

- Customer decisions

4.3 Dashboards and Data Visualization

- Why visualization matters
- Charts and graphs for managers
- KPIs and performance indicators
- Reading dashboards
- Avoiding misleading visualizations

4.4 Data Storytelling

- Converting numbers into meaningful business narratives
- Identifying the key message
- Communicating insights to decision-makers
- Combining data with managerial judgment

Class Activity:
Students interpret a sample business dashboard and present **three insights and two managerial actions**.

MODULE V – APPLICATIONS OF BIG DATA ACROSS MANAGEMENT FUNCTIONS

3 Hours

This module connects Big Data and Analytics directly with management education.

5.1 Marketing & Customer Analytics

- Customer segmentation
- Consumer behaviour
- Personalization
- Digital marketing analytics
- Customer sentiment
- Recommendation systems

5.2 Human Resource Management

- Workday & Other Applications Understanding